

A PROPOSAL TO DEVELOP SPMVV STUDENT SATELLITE -ELECTRICAL ENGINEERING DESIGN ASPECTS

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ABSTRACT:

The electrical engineering aspect of student CubeSat development focuses on the Electrical Power System (EPS), communications, and integration of power distribution to support all subsystems reliably in Low Earth Orbit (LEO). The EPS typically includes high-efficiency solar panels (e.g., deployable arrays generating at least 2.5 W for a 1U, using triple-junction cells) mounted on body or extendable surfaces, paired with rechargeable lithium-ion batteries (e.g., 18650 cells with UL 1642/2054 certification, capacity ≥ 10 Wh) for energy storage during eclipses. Power management employs Maximum Power Point Tracking (MPPT) controllers, DC-DC converters (e.g., 3.3V, 5V, 12V rails), and Battery Management Systems (BMS) to regulate voltage, prevent overcharge, and interface with the Command & Data Handling (C&DH) system. Safety features include multiple electrical inhibits like Remove Before Flight (RBF) pins and separation switches to delay activation post-deployment (e.g., no transmissions for 45 minutes). Communications involve radios, antennas (e.g., patch or monopole), and transceivers for telemetry downlink/uplink, often using amateur bands with Software-Defined Radios (SDR) for flexibility. EMI/EMC compliance is critical, adhering to MIL-STD-461F standards, with testing for harmonic suppression. Student designs prioritize COTS components for cost-efficiency, as seen in missions like ChargerSat-1 with deployable panels doubling power input. Challenges include balancing limited power budgets (e.g., 8W average) with payload demands, requiring efficient distribution and environmental testing (vibration, vacuum) to qualify for manned flights like ISS deployments.